ChatGPT

Project Plan: Predicting Environmental Exposure–Driven Child Health Risk in Bihar

Executive Summary: We propose an end-to-end ML project to identify Bihar districts at highest child health risk from environmental exposures. The model will predict a key child health outcome (e.g. under-5 stunting rate) using data on groundwater contaminants (arsenic, fluoride, etc.), precipitation/flood patterns, and socio-demographics (population density, sanitation, etc.). This goes beyond routine government reports by **integrating multi-sector data** into a predictive, explainable framework and prioritizing emerging hotspots and intervention targets. The final deliverables will include a clean merged dataset, an explainable ML model, risk maps, and clear policy recommendations with quantification of uncertainty.

Objectives

- **Estimate environmental exposure risk:** Predict district-level child health risk (chosen outcome: under-5 stunting prevalence) as a function of groundwater quality, rainfall/flood patterns, and socio-demographics.
- **Identify drivers:** Use explainable ML (e.g. SHAP) to discover which factors (arsenic, floods, sanitation, etc.) most influence the risk.
- **Produce actionable outputs:** Generate district risk maps and rank interventions. Emphasize where new interventions (e.g. water treatment, nutrition programs) would have greatest impact.

Novelty / Value Beyond Existing Reports

Government agencies *already map contaminated blocks*, but **no single framework** combines environment and health data to predict *future health burden*. This project's novelty is in **integrating diverse official datasets** and applying ML to forecast *health risk* rather than merely documenting contamination. For example, NFHS-5 shows 43% of Bihar's under-5 children are stunted [1], but current plans lack fine-grained exposure analysis. Our ML model can highlight *emerging hotspots* of risk (e.g. districts with rising contamination or worsening sanitation) *before* health surveys catch up. By focusing on a biological outcome (child nutrition/health), the project clearly serves an **environmental biotechnology/public health** purpose and not just a data exercise.

Chosen Target Outcome

We select **under-5 stunting prevalence** (percent of children 6–59 months stunted) at the district level as the target variable. *Justification:* Stunting is a robust indicator of chronic undernutrition linked to repeated infections and poor sanitation/water. It is measured in NFHS-5 (2019–21) for all Bihar districts [1]. High stunting (43% statewide) signals serious child health risk [1]. Predicting stunting focuses on biological impact (nutrition/health) rather than just contamination presence, giving a clear biotech/public-health angle. Other candidates (e.g. child anemia) could be used, but stunting directly reflects cumulative environmental and nutritional exposures.

Data Sources and Variables

We will use **one authoritative source per variable**. The main inputs are summarized below:

Variable / Feature	Data Source (Single Authoritative)	URL / Reference (in English)	Key Fields	Years Covered
Under-5 stunting rate	NFHS-5 (2019–21) *Bihar District Fact Sheet* (MoHFW/IIPS) [1]	MoHFW NFHS-5 Bihar Report (May 2021)	District ID, % of children 6–59m stunted	2019–2021
Groundwater arsenic (μg/L)	CGWB Bihar Groundwater Quality Yearbook (2022–23)	CGWB Publications (Bihar Yearbook)	District, As (mean/median, #samples)	2022–2023
Groundwater fluoride (mg/L)	CGWB Bihar Groundwater Quality Yearbook (2022–23)	CGWB Publications (Bihar Yearbook)	District, F (mean/max, #samples)	2022–2023
Groundwater nitrate (mg/L)	CGWB Bihar Groundwater Quality Yearbook (2022–23)	CGWB Publications (Bihar Yearbook)	District, NO_3 (mean/max, #samples)	2022–2023
Annual monsoon rainfall (mm)	India Meteorological Dept (gridded daily rainfall) via Bhuvan/ IMD	IMD Pune Gridded Rainfall (0.25°)	District annual total (JJA rain)	2010–2023
Population density (per km²)	Census of India 2011 (district level) [2]	Census 2011 District Abstract (Bihar)	District, pop, area	2011 (projected to 2020)
Open-defecation rate (%)	Census 2011, Household Amenities Table	Census 2011 (DCHB Bihar)	District, % HHs practicing open defecation	2011
Sanitation coverage	NFHS-5 (2019–21) *Bihar District Fact Sheet* [3]	MoHFW NFHS-5 Bihar Report	District, % HHs with toilet/ latrine	2019–2021
Other socio-demo features	Census 2011 (district) or NFHS-5	Census Bureau/ NFHS report	District, literacy rate, female pop.	2011 / 2019–21

Notes:

- CGWB Yearbook (Bihar) is the authoritative source for groundwater contaminants [4] [5].
- NFHS-5 Bihar report is the single source for all child health and sanitation indicators [3] [1].
- Rainfall will come from IMD's official gridded dataset (via download) or published summaries; e.g. Bihar Climate Center reports yearly and seasonal anomalies [6] [7].

Data Download Links and Files:
We will gather these sources as follows (summarized, with example commands):

- **CGWB Yearbook PDF:** (Manual PDF). Obtain from CGWB site. For example:

```
wget -O data/Bihar_GroundwaterYearbook2023.pdf \
     https://cgwb.gov.in/.../Bihar_GroundWaterYearbook2022-23.pdf
```

Fields: Contains tables of As, F, NO_3 etc. by district/well (manual extraction required).

- **NFHS-5 Bihar Fact Sheet:** (Manual PDF). Download from DHS program or MoHFW:

```
wget -O data/NFHS5_Bihar_FactSheet.pdf \
     https://ruralindiaonline.org/pdf/Bihar-NFHS5-Report.pdf
```

Fields: Tables for each district's stunting %, open-defecation %, etc. (manual or PDF-to-CSV).

- **IMD Gridded Rainfall:** (Access via online portal/API). For example, use Python to fetch from NOAA/IMD:

```python
import cdsapi
c = cdsapi.Client()
c.retrieve('ecad-rain',
{'variable':'tp','format':'zip','product_type':'reanalysis','year':
['2010','2011',...,'2023'],'area':[27.5,83.5,24.5,88.5],'month':
['06','07','08','09']}, 'data/IMD_rainfall_2010-23.zip')
```

Fields: Unzip to get daily NetCDF; aggregate to yearly totals per district.

- **Census 2011 District Data:** Public PDFs/CSVs:

```
wget -O data/Census_Bihar2011_DistrictData.zip \
     http://censusindia.gov.in/.../DCHB_Bihar.zip
```

Fields: District population, area, literacy, %households with toilets (from Household Table).

- **NFHS-5 / Census for Sanitation:** Sanitation (% households with latrine) from NFHS-5 Fact Sheet or Census 2011 (DCHB, Household amenities).
- If any dataset is not directly available (e.g. gridded IMD), we will use the **best single alternative** (e.g. CHIRPS rainfall if IMD unavailable, or district nutrition profiles from NITI if NFHS raw data not accessible).

All data will be placed in `/data` subfolders by type (e.g. `/data/raw_groundwater`, `/data/raw_health`, etc.). Exact file names and URLs will be documented. Most sources require manual PDF extraction.

Required Input Schema and Data Dictionary

We will merge data at the **district-year** level (district as key). Example schema for the analysis table:

Field Name	Description	Type
`State`	Always "Bihar"	string
`District`	District name (2011 boundaries)	string
`Year`	Calendar year (for time-varying data)	integer
`Monsoon_Rainfall_mm`	Total Jun–Sep rainfall (mm) in district	float
`Arsenic_ppb`	Mean arsenic concentration (μg/L) in groundwater	float
`Fluoride_mgL`	Mean fluoride concentration (mg/L) in groundwater	float
`Nitrate_mgL`	Mean nitrate concentration (mg/L) in groundwater	float
`OpenDefec_pct`	% households practicing open defecation	float
`ToiletCoverage_pct`	% households with latrine/sanitation facilities	float
`Population`	District population	integer
`Area_sqkm`	District area (sq. km)	float
`PopDensity_perSqkm`	Derived: population density	float
`OtherVars...`	(e.g. literacy rate, % urban)	…
`Stunting_pct`	**Target:** % children 6–59m who are stunted	float

All input variables (rainfall, arsenic, etc.) will be aligned to the same year or closest prior year of the outcome (e.g. use 2020 groundwater vs 2019–21 NFHS). The data dictionary (descriptions, units, source) will be compiled as a separate deliverable.

ETL and Data Processing Steps

1. **Extract raw data:** Load PDFs/CSVs as dataframes or tables:

2. Parse CGWB tables (e.g. via Tabula or manual copying) into CSV columns per district.

3. Read NFHS fact sheets to get health and sanitation indicators per district.
4. Read IMD rainfall CSV/NetCDF; aggregate to district totals (using district shapefile).

5. Read Census tables for population, area, sanitation.

6. **Cleaning & Standardization:**

7. **Standardize names:** Ensure district names match across sources (e.g. "Pashchim Champaran" vs "West Champaran"). Use a lookup or census district codes to unify.

8. **Units:** Convert all concentrations to standard units (µg/L for arsenic, mg/L for others) if needed. Convert population to absolute and density.
9. **Missing values:** Mark missing if a district has no sample (CGWB) or no data (small districts). We will retain districts with partial data but note missingness for modeling.
10. **Outliers:** Check for impossible values (e.g. negative rainfall) and cap or mark them.
11. **Temporal alignment:** NFHS outcome is a multi-year survey (2019–21). We will assign each district's stunting value to year 2020 as an average year. All other inputs will be from 2018–2020 as possible. For time-series features like rainfall, include 5-year averages or trends up to 2020.
12. **Merge datasets:** Join on District and Year. If some variables are static (Census 2011), they will repeat for all years or be forward-filled to 2020.
13. **Final cleaning:** Verify no mismatches or duplicates. Save the merged table as the analysis dataset (`.csv`).

Feature Engineering

Potential features to capture exposure patterns and trends:

- **Static geographics:** District area, elevation or floodplain indicator (if available from DEM).
- **Temporal trends:**
- 3-year moving average of rainfall or number of extreme rain events.
- Change in groundwater level or contamination over recent years (if multi-year data available).
- **Seasonal signals:**
- Separate pre-monsoon vs post-monsoon groundwater depth or rainfall.
- **Sanitation/WASH:**
- % households with safe water source (from NFHS/Census).
- Combine open defecation and sanitation into a "Sanitation Index".
- **Demographics:**
- Population density (pop/area).
- Child population share (if available).
- Rural vs urban % (from census).
- **Interaction terms:** e.g. (rainfall × open defecation %) to capture flooding + poor sanitation risk.
- **Previous Outcome:** If using NFHS-4 (2015–16) stunting, include that to predict NFHS-5 (not required if modeling single cross-section).

Each engineered feature will be documented. Categorical variables (if any) like agro-climate zone can be one-hot encoded.

Modeling Approach

- **Type:** We will frame this as a regression predicting *continuous* stunting rate, or as a classification (e.g. high-risk vs low-risk based on a threshold). Primarily regression with R^2/MAE is straightforward.
- **Algorithms:**
- Start with a simple baseline (Linear Regression).
- Tree-based ensemble: Random Forest or XGBoost/LightGBM for non-linear relationships. These handle mixed numeric data and missing values well.

- We may test a simple spatial model (e.g. geostatistical kriging of contamination) if warranted, but focus on explainable models.
- **Hyperparameters:**
- For tree models, tune number of trees (e.g. 100–1000), max depth (3–10), learning rate (if boosting).
- Use cross-validation to pick parameters (e.g. grid search with 5×5 folds over parameter grid).

Train/Test Split and Validation

- **Spatial split:** Perform **leave-one-district-out** validation or clustered CV to ensure the model generalizes to unseen districts. For example, train on 35 districts, test on 3 (rotated).
- **Temporal split:** If we incorporate NFHS-4 (2015–16) and NFHS-5, we can train on NFHS-4 data (2015) and validate on NFHS-5 (2020) to simulate forecasting. If only one time point, we simply use cross-validation on districts.
- **Cross-validation:** 5-fold spatial CV (group by district) plus a holdout of the entire year (if multi-year).

Evaluation Metrics

- **Regression metrics:** RMSE (root mean square error), MAE, R^2.
- **Classification metrics (if thresholding risk):** F1-score, precision/recall, ROC-AUC (for high-risk vs low-risk). *We will emphasize recall/sensitivity for the "high-risk" class.* That is, it is more critical to catch all districts at risk of high stunting than to be perfectly accurate on safe ones.
- **Calibration:** Check predicted vs actual distribution of stunting.
- **Comparison to baseline:** Also check a naive baseline (e.g. mean prediction, or ranking by arsenic alone) to quantify model gain.

Minimum acceptable performance will be set (e.g. $R^2 > 0.5$ or recall > 0.8 for high-risk class) based on cross-val results.

Explainability and Interpretation

- **Feature Importance:** Use SHAP values to rank features (arsenic, rainfall, sanitation, etc.) by their impact on predictions [3] [7].
- **Partial Dependence / Accumulated Local Effects:** Show how risk changes with key variables (e.g. risk vs arsenic level).
- **Maps:** Plot district-level actual vs predicted risk on Bihar map, highlighting hotspots.
- **Trend Analysis:** Identify "emerging hotspots" by finding districts with predicted risk rise (if using a temporal model) or ones heavily influenced by worsening exposures.

Example interpretation: *"We find that predicted stunting risk is highest in north-central Bihar, driven by a combination of high arsenic and low sanitation. SHAP analysis shows that district arsenic levels and open-defecation rates contribute most to the risk score."*

Uncertainty Quantification

- **Prediction intervals:** For tree models, we can use quantile regression forests or bootstrapping to get prediction intervals for each district.
- **Uncertainty maps:** Produce a "confidence" map showing where predictions are most uncertain (e.g. due to sparse data).

- **Data gaps:** Document districts with very few groundwater samples or missing health data; highlight these as uncertainty sources.
- **Model robustness:** Test how sensitive predictions are to slight data changes (e.g. plus/minus contamination values).

In the report, we will explicitly discuss limitations (e.g. no individual-level data, potential misalignment of years, etc.) and quantify how much uncertainty they introduce.

Deliverables

We will produce the following artifacts:

Deliverable	Description / Format
Clean merged dataset	CSV/Parquet with all features and target per district-year (incl. codebook)
Data dictionary	Markdown or PDF listing each field, units, source
ETL pipeline code	Well-documented Python/R scripts or Jupyter notebooks for data extraction and cleaning
Modeling code & model artifacts	Code for training final model (Jupyter/Python), plus saved model file (pickle)
Model evaluation report	Table of metrics, CV results, residual analysis
Explainability analysis	SHAP plots, feature importance charts, partial-dependence plots
Risk maps	(a) Predicted stunting risk by district map, (b) Uncertainty map (e.g. CI width)
Intervention priority list	Top-10 districts by risk (with any recommended actions)
Policy brief / presentation slides	Summary of findings, methods, and recommendations
Full written report	Scientific report / thesis chapter covering context, methods, results, discussion
Final Abstract	Concise summary of the entire study for stakeholders

Most deliverables will be digital (CSV, PDF, PNG) and organized in a logical folder structure (e.g. `/data`, `/scripts`, `/results`, `/reports`).

Example folder structure:

```
/Bihar_HealthRisk_Project
│
├── data/
│   ├── raw_groundwater/       # downloaded CGWB PDFs, extracted CSVs
│   ├── raw_climate/           # rainfall NetCDF or CSV
│   ├── raw_health/            # NFHS-5 PDF, extracted stats
│   ├── raw_census/            # Census Excel/PDFs for Bihar
```

```
│   └── processed/            # cleaned combined CSVs
│
├── notebooks/                # Jupyter notebooks for ETL and modeling
├── src/                      # Python scripts or modules
├── results/
│   ├── figures/              # maps, plots (PNG)
│   └── models/               # saved model files, SHAP values
│
├── outputs/
│   ├── report.pdf            # final report/thesis
│   └── presentation.pptx     # slides
│
└── requirements.txt          # dependencies and versions
```

Computing Environment:
A standard data science environment is sufficient. We recommend Python (3.8+) with packages: `pandas`, `numpy`, `scikit-learn`, `xgboost`, `shap`, `geopandas`, `rasterio`, etc. GPU is not needed. All dependencies will be listed in `requirements.txt` with versions. The workflow is reproducible (scripts or notebooks documented step-by-step).

Implementation Timeline

```
gantt
    dateFormat  YYYY-MM-DD
    title       Bihar Health Risk ML Project Timeline

    section Data Collection & ETL
    Download and extract CGWB data      :done,     des1, 2026-05-01,
2026-05-10
    Download NFHS & Census data         :done,     des2, 2026-05-05,
2026-05-15
    Fetch and process rainfall data     :active,   des3, 2026-05-10,
2026-05-25
    Clean and merge datasets            :         des4, 2026-05-20,
2026-06-05

    section Feature Engineering
    Create derived features             :         feat1, 2026-06-06,
2026-06-12
    Exploratory data analysis           :         feat2, 2026-06-06,
2026-06-15

    section Modeling
    Baseline and model training         :         mod1, 2026-06-13,
2026-06-25
    Cross-validation and tuning         :         mod2, 2026-06-26,
2026-07-05
    Select best model                   :         mod3, 2026-07-06,
2026-07-08
```

```
    section Evaluation & Visualization
    Compute metrics / interpret        :         eval1, 2026-07-09,
2026-07-15
    Generate maps and charts           :         eval2, 2026-07-16,
2026-07-22
    Uncertainty analysis               :         eval3, 2026-07-18,
2026-07-25

    section Reporting
    Write draft report                :         rep1, 2026-07-23, 2026-08-01
    Prepare presentation slides        :         rep2, 2026-08-02,
2026-08-05
    Review and finalize deliverables   :         rep3, 2026-08-06,
2026-08-10
```

Legend: Dates are illustrative (month/day-year).

Summary

This comprehensive ML project will integrate CGWB water quality data, IMD rainfall, and health/socioeconomic statistics to **predict child stunting risk** in Bihar. It emphasizes scientific rigor (clear data provenance, reproducibility) and policy relevance (risk maps, intervention targets). By focusing on exposure **to health outcomes**, it goes beyond mapping contamination to highlighting where *child health* is most threatened by the environment. The model's explainable insights and uncertainty quantification will provide reliable guidance to both scientists and policymakers.

Required inputs from user: None beyond what is listed in the data sources table. All data will be obtained from the cited official sources. If any data (e.g. certain PDFs or APIs) require manual download or credentials, we will note that in the pipeline instructions. The user simply needs to run the provided scripts or use the wget commands to fetch the data as documented.

Citations: Authoritative references used include Bihar's NFHS-5 report [1] [3], India Meteorological Dept data via Bihar climate summaries [6] [7], and CGWB groundwater reports (for content context) [4] [5].

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[1] [3] National Family Health Survey (NFHS-5) 2019-21: Bihar
https://ruralindiaonline.org/bn/library/resource/national-family-health-survey-nfhs-5-2019-21-bihar/

[2] dse.bihar.gov.in
http://dse.bihar.gov.in/Source/Provisional%20Population%20Totals%202011-Bihar.pdf

[4] [PDF] Ground Water Year Book,Bihar - CGWB
https://cgwb.gov.in/cgwbpnm/public/uploads/documents/1703219758358094014file.pdf

[5] Arsenic exposure in Indo Gangetic plains of Bihar causing increased ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC7841152/

[6] [7]
https://forestonline.bihar.gov.in/SKMCCC/Rainfall.aspx